



CIMA STRATEGIC CASE STUDY

Mock Exam 1

Rotomyne

Unseen

Task 1 - Unseen case material

You receive the following email from the Chief Finance Officer (CFO).

From: Martin Jacobs, CFO

To: Senior Manager (SM)

Subject: New brine mine planning permission and financing

Hi SM,

A few months ago, a large brine reservoir was discovered deep below an arid area in the South American country Dryland, which is known for being politically and economically unstable. The country's government subsequently invited tenders for the mining of lithium from the reservoir. We applied and, after a tough tendering process, were awarded the mining licence. Our success was mainly due to our experience in brine mining.

Last month, we submitted the planning permission to construct the new brine mine. However, the mine's planning permission was held up as the local government planning office received complaints that the mine would pump excessive amounts of brine, leading to freshwater depletion. Although we refuted this, a recent study has supported the complaints, and local protests are now planned (see the attached email). We had reached agreements in principle with our bank to fund this P\$50-million mining project and with a local construction firm for the building of the brine mine. However, both are nervous due to negative media coverage. The bank may even withdraw project funding.

So that I can discuss this with the CEO, I need your input as follows:

- Could you **recommend some safeguards** that we can put in place to retain the support of our **bank and construction firm as well as local community protesters**?

[Sub task (1a) - 55%]

- Could you **discuss** the factors we should consider when choosing between equity and debt as a source of finance for the project and **recommend, with a detailed justification**, the preferred type of finance?

[Sub task (1b) - 45%]

Thanks,
Martin

Reference material

Email to the CFO from the Non-Executive Chair

From: Iresh Jayawardena, Non-Executive Chair

To: Martin Jacobs, CFO

Subject: Environmental protests

Hi Martin,

I have been contacted by the head of a local community in Dryland. She has made me aware of a recent scientific study that was conducted in a neighbouring country which has brine mines for lithium extraction. The study has shown that a decrease in brine reservoirs also corresponds to a decrease in the freshwater water table.

She said that if this were to occur as a result of our proposed brine mine, this would affect freshwater levels in wells in local villages. As such, a residents' group is in the process of organising a peaceful, non-violent protest at our planned construction site.

This is potentially a major risk. We need to prepare for these protests and limit any damage to property, equipment and personnel. We have our own onsite security, but we would need to bolster this by hiring security guards from a private security company. Police would also need to be made aware of the planned protests.

Additionally, we have a duty of care to the local community, and we respect their right to protest. As such, we also need to protect the protesters too, as they will be exposed to high temperatures (en route to and at the construction site), and there may be scuffles with our onsite security and the local police (who have a reputation for being heavy-handed)!

Best regards,
Iresh

Task 2 - Unseen case material

The following email is waiting for you when you arrive at the office.

From: Martin Jacobs, CFO

To: Senior Manager (SM)

Subject: DLE plant proposal and government relations

Hi SM,

It has been a few weeks, and the protests have yet to happen. The Drylandian government is up for re-election soon, so it wants to gain support from local communities affected by lithium mining by addressing their freshwater concerns. As such, it has now ruled out the use of a traditional brine mine. The licence for lithium extraction will now be awarded based on the direct lithium extraction (DLE) method, which it sees as a better method.

I discussed this with the Production Director (Andrey) and summarised our findings in the attached briefing note. In summary, a DLE plant has less environmental impact. Also, the hot brine reservoir could be used to power the DLE plant via geothermal electricity generation and thus operate without carbon emissions.

However, it is expensive, and we have no experience in DLE. That said, I feel we must take the opportunity to build a DLE plant, especially as governments worldwide are hesitant to grant permission to extract lithium due to the environmental damage caused by hard rock and brine mines. Plus, it is difficult to find suitable sites for DLE. We will obviously have a lot of competition for the DLE licence. To increase our chances of being awarded the DLE licence, we have to be in the government's good books.

I need an email from you to support my follow-up meeting with the CEO:

- Could you analyse the DLE plant opportunity for its suitability, acceptability, and feasibility, as well as provide a justified recommendation and action plan?

[Sub task (2a) - 55%]

- Could you also discuss and recommend suitable approaches to managing our relationship with the government?

[Sub task (2b) - 45%]

Many thanks,

Martin

Reference material

Briefing note - Lithium site: environmental impact and financials

Criteria	Traditional brine mine	Direct lithium extraction plant
Current number in our portfolio	3	0
Environmental impact		
Carbon dioxide emissions per tonne of lithium	5,000kg	0kg (geothermal source)
Fresh water used per tonne of lithium	500,000 litres	3,000 litres
Use of land per tonne of lithium (annual capacity)	3,200 square metres	1 square metre
Power source	Electricity (fossil-based)	Electricity (geothermal)
Extraction of lithium from brine approach	Evaporation	Solvent, absorption or ion exchange
Recovery of lithium from brine	30-50%	70 - 90%
Financial analysis		
Finance cost	10%	10%
Investment	P\$50m	P\$100m
NPV	P\$50m	P\$20m
Time taken to produce lithium carbonate	12-18 months	3 months

A geological survey of the mining site indicated that the brine temperature is, on average, 55 degrees centigrade (131 degrees Fahrenheit).

Task 3 - Unseen case material

You have just received the following email from the CFO.

From: Martin Jacobs, CFO

To: Senior Manager (SM)

Subject: JV governance, political risk management, and sustainability training

Hi SM,

Thanks for the work on the proposal and government relations. We are inexperienced with the direct lithium extraction (DLE) method, and Dryland is unfamiliar to us. Our Board is also busy, so a joint venture (JV) with a more experienced company may help us. I'd suggest a 50:50 JV split of control, costs and profit. To effectively manage our strategic risks, the JV needs good governance. (N.B. Porrrland's Corporate Governance Code mirrors the UK Corporate Governance Code.)

- Could you discuss the JV's corporate governance requirements regarding non-executive directors and committees and how to fulfil such requirements?

[Sub task (3a) - 60%]

Also, it might be prudent to enter a JV specifically with a Drylandian company. This may improve our chances of securing and keeping hold of the DLE licence, especially since Dryland is politically unstable and may also change government in the upcoming election.

- Could you discuss ways a local company JV would help manage political risks?

[Sub task (3b) - 20%]

Lastly, the freshwater concerns (a sustainability issue) would be 'technically' addressed by the DLE plant. However, to adequately staff the project, the HR director believes staff need sustainability training, but the Production Director disagrees (see attachment).

- Could you evaluate the need for sustainability training and provide a clear recommendation?

[Sub task (3c) - 20%]

I'll need the information in a briefing note for a Board meeting.

Many thanks,
Martin

Reference material

Email attachment - Extract of the Board's discussion on training

CEO (Shaista)	"The DLE plant project proposal is progressing well. The DLE method would help resolve the freshwater issues."
HR Director (Cecilia)	"The Drylandian local community's freshwater concerns highlight the need for greater ethical and social consciousness for staff that would work on the DLE plant project and for staff on other projects."
Chair (Iresh)	"Yes, Corporate Social Responsibility (CSR) does seem to be a running theme across all our projects."
HR Director (Cecilia)	"I would suggest a company-wide external 2-day course on sustainability, aimed at 100 key staff (including directors and management). It would cost P\$1,500 per person per day."
Production Director (Andrey)	"This would be a waste of money as sustainability was irrelevant to most operational managers as well as requiring staff to be away from work. The disruption would cost P\$125 per person per day."
HR Director (Cecilia)	"This just shows that CSR, of which sustainability is an aspect, is not being taken seriously enough! This skills gap could threaten the success of the potential DLE plant project as well as other projects as our staff could inadvertently upset key stakeholders."
FD (Martin)	"Training costs and non-financial considerations probably need to be weighed up."

Present value tables

Present value of US\$1 i.e. $(1 + r)^{-n}$ where r = interest rate, n = number of periods until payment or receipt.

Period s (n)	Interest rates (r)									
	1%	2%	3%	4%	5%	6%	7%	8%	9%	10%
1	.990	.980	.971	.962	.962	.943	.935	.926	.917	.909
2	.980	.961	.943	.925	.907	.890	.873	.857	.842	.826
3	.971	.942	.915	.889	.864	.840	.816	.794	.772	.751
4	.961	.924	.888	.855	.823	.792	.763	.735	.708	.683
5	.951	.906	.863	.822	.784	.747	.713	.681	.650	.621
6	.942	.888	.837	.790	.746	.705	.666	.630	.596	.564
7	.933	.871	.813	.760	.711	.665	.623	.583	.547	.513
8	.923	.853	.789	.731	.677	.627	.582	.540	.502	.467
9	.914	.837	.766	.703	.645	.592	.544	.500	.460	.424
10	.905	.820	.744	.676	.614	.558	.508	.463	.422	.386
11	.896	.804	.722	.650	.585	.527	.475	.429	.388	.350
12	.887	.788	.701	.625	.557	.497	.444	.397	.356	.319
13	.879	.773	.681	.601	.530	.469	.415	.368	.326	.290
14	.870	.758	.661	.577	.505	.442	.388	.340	.299	.263
15	.861	.743	.642	.555	.481	.417	.362	.315	.275	.239
16	.853	.728	.623	.534	.458	.394	.339	.292	.252	.218
17	.844	.714	.605	.513	.436	.371	.317	.270	.231	.198
18	.836	.700	.587	.494	.416	.350	.296	.250	.212	.180
19	.828	.686	.570	.475	.396	.331	.277	.232	.194	.164
20	.820	.673	.554	.456	.377	.312	.258	.215	.178	.149

Period s (n)	Interest rates (r)									
	11%	12%	13%	14%	15%	16%	17%	18%	19%	20%
1	.901	.893	.885	.877	.870	.862	.855	.847	.840	.833
2	.812	.797	.783	.769	.756	.743	.731	.718	.706	.694
3	.731	.712	.693	.675	.658	.641	.624	.609	.593	.579
4	.659	.636	.613	.592	.572	.552	.534	.516	.499	.482
5	.593	.567	.543	.519	.497	.476	.456	.437	.419	.402
6	.535	.507	.480	.456	.432	.410	.390	.370	.352	.335
7	.482	.452	.425	.400	.376	.354	.333	.314	.296	.279
8	.434	.404	.376	.351	.327	.305	.285	.266	.249	.233
9	.391	.361	.333	.308	.284	.263	.243	.225	.209	.194
10	.352	.322	.295	.270	.247	.227	.208	.191	.176	.162
11	.317	.287	.261	.237	.215	.195	.178	.162	.148	.135
12	.286	.257	.231	.208	.187	.168	.152	.137	.124	.112
13	.258	.229	.204	.182	.163	.145	.130	.116	.104	.093
14	.232	.205	.181	.160	.141	.125	.111	.099	.088	.078
15	.209	.183	.160	.140	.123	.108	.095	.084	.074	.065
16	.188	.163	.141	.123	.107	.093	.081	.071	.062	.054
17	.170	.146	.125	.108	.093	.080	.069	.060	.052	.045
18	.153	.130	.111	.095	.081	.069	.059	.051	.044	.038
19	.138	.116	.098	.083	.070	.060	.051	.043	.037	.031
20	.124	.104	.087	.073	.061	.051	.043	.037	.031	.026

Cumulative present value tables

This table shows the Present Value of US\$1 per annum, Receivable or Payable at the end of each year for n years.

$$\frac{1 - (1+r)^{-n}}{r}$$

Periods (n)	Interest rates (r)									
	1%	2%	3%	4%	5%	6%	7%	8%	9%	10%
1	0.990	0.980	0.971	0.962	0.952	0.943	0.935	0.926	0.917	0.909
2	1.970	1.942	1.913	1.886	1.859	1.833	1.808	1.783	1.759	1.736
3	2.941	2.884	2.829	2.775	2.723	2.673	2.624	2.577	2.531	2.487
4	3.902	3.808	3.717	3.630	3.546	3.465	3.387	3.312	3.240	3.170
5	4.853	4.713	4.580	4.452	4.329	4.212	4.100	3.993	3.890	3.791
6	5.795	5.601	5.417	5.242	5.076	4.917	4.767	4.623	4.486	4.355
7	6.728	6.472	6.230	6.002	5.786	5.582	5.389	5.206	5.033	4.868
8	7.652	7.325	7.020	6.733	6.463	6.210	5.971	5.747	5.535	5.335
9	8.566	8.162	7.786	7.435	7.108	6.802	6.515	6.247	5.995	5.759
10	9.471	8.983	8.530	8.111	7.722	7.360	7.024	6.710	6.418	6.145
11	10.368	9.787	9.253	8.760	8.306	7.887	7.499	7.139	6.805	6.495
12	11.255	10.575	9.954	9.385	8.863	8.384	7.943	7.536	7.161	6.814
13	12.134	11.348	10.635	9.986	9.394	8.853	8.358	7.904	7.487	7.103
14	13.004	12.106	11.296	10.563	9.899	9.295	8.745	8.244	7.786	7.367
15	13.865	12.849	11.938	11.118	10.380	9.712	9.108	8.559	8.061	7.606
16	14.718	13.578	12.561	11.652	10.838	10.106	9.447	8.851	8.313	7.824
17	15.562	14.292	13.166	12.166	11.274	10.477	9.763	9.122	8.544	8.022
18	16.398	14.992	13.754	12.659	11.690	10.828	10.059	9.372	8.756	8.201
19	17.226	15.679	14.324	13.134	12.085	11.158	10.336	9.604	8.950	8.365
20	18.046	16.351	14.878	13.590	12.462	11.470	10.594	9.818	9.129	8.514
Periods (n)	Interest rates (r)									
	11%	12%	13%	14%	15%	16%	17%	18%	19%	20%
1	0.901	0.893	0.885	0.877	0.870	0.862	0.855	0.847	0.840	0.833
2	1.713	1.690	1.668	1.647	1.626	1.605	1.585	1.566	1.547	1.528
3	2.444	2.402	2.361	2.322	2.283	2.246	2.210	2.174	2.140	2.106
4	3.102	3.037	2.974	2.914	2.855	2.798	2.743	2.690	2.639	2.589
5	3.696	3.605	3.517	3.433	3.352	3.274	3.199	3.127	3.058	2.991
6	4.231	4.111	3.998	3.889	3.784	3.685	3.589	3.498	3.410	3.326
7	4.712	4.564	4.423	4.288	4.160	4.039	3.922	3.812	3.706	3.605
8	5.146	4.968	4.799	4.639	4.487	4.344	4.207	4.078	3.954	3.837
9	5.537	5.328	5.132	4.946	4.772	4.607	4.451	4.303	4.163	4.031
10	5.889	5.650	5.426	5.216	5.019	4.833	4.659	4.494	4.339	4.192
11	6.207	5.938	5.687	5.453	5.234	5.029	4.836	4.656	4.486	4.327
12	6.492	6.194	5.918	5.660	5.421	5.197	4.968	4.793	4.611	4.439
13	6.750	6.424	6.122	5.842	5.583	5.342	5.118	4.910	4.715	4.533
14	6.982	6.628	6.302	6.002	5.724	5.468	5.229	5.008	4.802	4.611
15	7.191	6.811	6.462	6.142	5.847	5.575	5.324	5.092	4.876	4.675
16	7.379	6.974	6.604	6.265	5.954	5.668	5.405	5.162	4.938	4.730
17	7.549	7.120	6.729	6.373	6.047	5.749	5.475	5.222	4.990	4.775
18	7.702	7.250	6.840	6.467	6.128	5.818	5.534	5.273	5.033	4.812
19	7.839	7.366	6.938	6.550	6.198	5.877	5.584	5.316	5.070	4.843
20	7.963	7.469	7.025	6.623	6.259	5.929	5.628	5.353	5.101	4.870